



SCOPE

DA2 - Project Review II

- Course Code: BCSE302L
- Course Title: Database Systems
- Class Number: CH2025260501233 / CH2025260501238
- Semester: Winter 2025-26
- Slot: A1+TA1 / A2+TA2
- Course Faculty: Dr. L.M. Jenila Livingston

Rubrics:

Description	Marks
Normalization	3
Back-end Design	4
Viva/Quiz	3

Objective: Students will:

- Convert ER-mapped tables into well-structured relations
- Apply normalization (1NF → 2NF → 3NF → BCNF)
- Design a clean backend database schema ready for implementation

Input (From Review-1 Work)

You have already completed:

- ER Diagram
- ER-to-Relational Mapping (Tables)

Use the same tables for this assignment. If you have added any new tables, you may use the updated version—just make sure to mention it

Part 1: Normalization Process

For **each table**, perform the following steps:

Step 1: First Normal Form (1NF)

- **Insert any five records** for the tables and ensure:
 - o No repeating groups
 - o All attributes are atomic
- If violations exist:
 - o Convert into separate rows/tables

Step 2: Second Normal Form (2NF)

- Identify:
 - **Candidate Keys**
 - **Partial Dependencies**
- Action:
 - Remove partial dependency
 - Create new tables if needed

Step 3: Third Normal Form (3NF)

- Identify:
 - **Transitive dependencies**
- Action:
 - Remove non-key dependencies
 - Split tables accordingly

Step 4: Boyce-Codd Normal Form (BCNF)

- Check:
 - For every functional dependency $X \rightarrow Y$, **X must be a super key**
- If violated:
 - Decompose the table

Expected Output for Each Table

For **each relation**, students must present:

1. **Original Table**
2. **Functional Dependencies (FDs)**
3. **Candidate Keys**
4. **1NF Table**
5. **2NF Decomposition**
6. **3NF Decomposition**
7. **BCNF Final Tables**
8. **Justification for each step**

Important Notes

- Clearly show **step-by-step decomposition**
- Avoid skipping directly to BCNF
- Use proper notation (e.g., $A \rightarrow B$)
- **If a table is already in a normal form, justify why**

Part 2: Backend Database Design

Create Final Schema

- List all final tables (BCNF)
- Define:
 - **All Constraints** including Primary Keys and Foreign Keys

Show Backend Implementation